CHAPTER 1
INTRODUCTION

1.1 PURPOSE OF THIS MANUAL

This manual was prepared to assist entry-level engineers in the design of air-conditioning systems. It is also usable—in conjunction with fundamental HVAC&R resource materials—as a senior- or graduate-level text for a university course in HVAC system design. This manual was intended to fill the void between theory and practice, to bridge the gap between real-world design practices and the theoretical knowledge acquired in the typical college course or textbook. Courses and texts usually concentrate on theoretical calculations and analytical procedures or they focus upon the design of components. This manual focuses upon applications.

The manual has two main parts: (1) a narrative description of design procedures and criteria organized into ten chapters and (2) six appendices with illustrative examples presented in greater detail.

The user/reader should be familiar with the general concepts of HVAC&R equipment and possess or have access to the four-volume *ASHRAE Handbook* series and appropriate ASHRAE special publications to obtain grounding in the fundamentals of HVAC&R system design. Information contained in the *Handbooks* and in special publications is referenced—but not generally repeated—herein. In addition to specific references cited throughout the manual, a list of general references (essentially a bibliography) is presented at the end of this chapter.

The most difficult task in any design problem is how to begin. The entry-level professional does not have experience from similar projects to fall back on and is frequently at a loss as to where to start a design. To assist the reader in this task, a step-by-step sequence of design procedures is outlined for a number of systems.

- Ensure that there is ready access for routine maintenance tasks, such as changing air filters.
- Specify accessible service shutoff drain and vent valves for all water coils and risers.
- Incorporate appropriate chemical treatment equipment for open- and closed-loop water systems.

2.5.3 Noise and Vibration Control

Noise and vibration from HVAC&R equipment can be reduced by proper selection of equipment, vibration control, and the interposition of sound attenuators and barriers. Residual sound may still be objectionable, however, when equipment is located near occupied areas. HVAC&R and architectural design decisions must go hand-in-hand since acoustical control may involve equipment location, floor and wall assemblies, and room function. HVAC&R equipment locations may influence spatial planning—so that areas requiring a low-noise environment are not located next to major or noisy equipment. The HVAC engineer should alert other members of the building design team regarding the location of noisy equipment. On large, acoustically sensitive buildings (e.g., theaters, museums), an acoustical consultant should be part of the design team.

Noise and vibration transmission to an occupied space by system components will be an important consideration in system selection and design. Even after a system has been selected, component selection will significantly affect system acoustical performance. Noise can be transmitted to occupied spaces from central station equipment along several airborne paths, through air or water flows, along the walls of ducts or pipes, or through the building structure. If central station pumps or fans are used, each of these paths must be analyzed and the transmission of sound and vibration reduced to an acceptable level. Supply air outlets (and other terminal devices) must be selected to provide appropriate acoustical performance.

An initial step in noise control is to establish noise criteria for all spaces. These criteria should be communicated to the client early in the design process. All noise-generating sources within the air-conditioning system must be identified. Then, the effects of those sources can be controlled by

- equipment selection,
- air and water distribution system design,
- structural/architectural containment,
- dissipative absorption along the sound path, and
- isolation from the occupied space, where feasible.

Design of piping from the tower sump to the pump should be undertaken with caution. Place the sump level above the top of the pump casing to provide positive prime. Pitch all piping up, either to the tower or to the pump suction, to eliminate air pockets. Suction strainers should be equipped with inlet and outlet gauges that indicate when cleaning is required. Normally, piping is sized to yield water velocities between 5 and 12 fps [1.5 and 3.7 m/s]. Use friction factors for aged or old pipes.

In cooling tower installations, ascertain that sufficient NPSH (net positive suction head—see Section 5.5.2) for the condenser-water pump is provided, and design the connections so that the flow through each cooling tower has the same pressure drop. Install equalizing lines of proper size between towers; otherwise, one tower will fill and the other one will overflow. Winterize the tower piping if necessary or provide for drain-down when pumps shut off. In the latter case, the isolation valves must be located in a heated space. Provide valving or sump pump isolation so that any single cell can be serviced while the others are active.

5.4.6 Evaporative Condensers

Evaporative condensers are used to reject heat to the outdoor air by means of the evaporation of water. Water is sprayed on the condenser coils, and the resultant heat of vaporization, combined with any sensible heat effect, cools the liquid inside the coils. The residual spray water is usually recirculated. An evaporative condenser uses much less makeup water than a cooling tower in a conventional water-cooled condenser arrangement (approximately 1.6 to 2.0 gph/ton [0.5 to 0.6 mL/s per kW] by evaporation plus 0.5 to 0.6 gph/ton [0.15 to 0.18 mL/s per kW] for bleed water). Evaporative condensers are used primarily as refrigeration system condensers. In such applications, they should be located as close as possible to the refrigeration machinery to keep the refrigerant lines short. In the winter, the water can often be shut off with the dry coil providing adequate heat rejection. Evaporative condensers are also used to cool recirculating water in closed-loop water-source heat pump systems (see Chapter 7).

Figure 5-8 shows two evaporative condenser types using city, well, or river makeup water. Piping should be sized in accordance with the principles outlined in Section 5.4.5 with flow velocities of 5 to 10 fps [1.5 to 3.0 m/s]. Where city water is used, a pump is usually not required since the water arrives under pressure. For well or river water, pumps may be necessary, in which case the